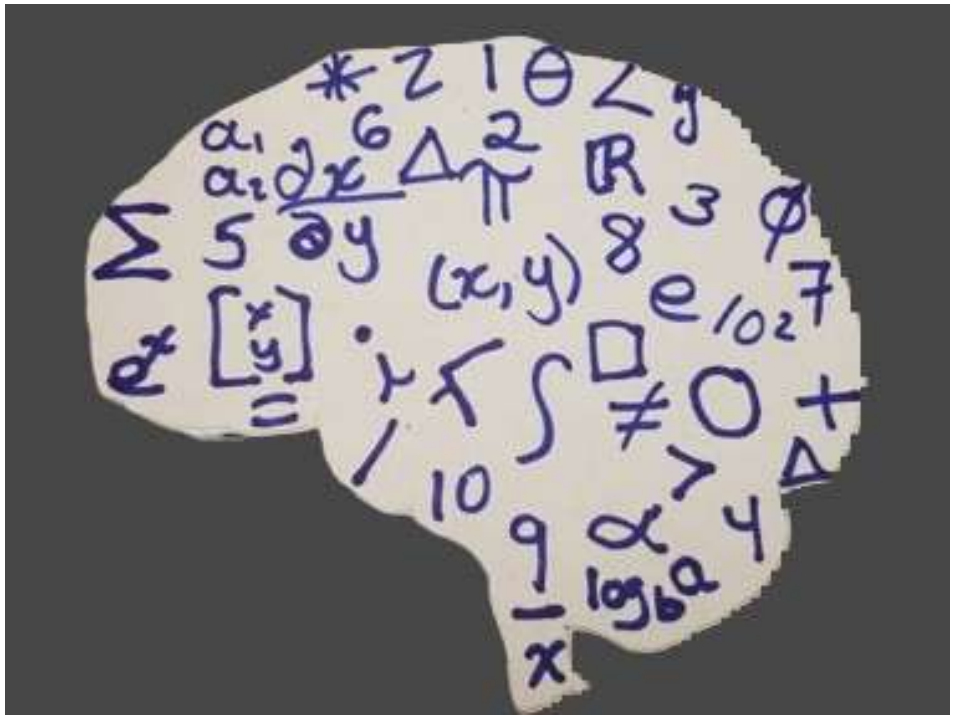
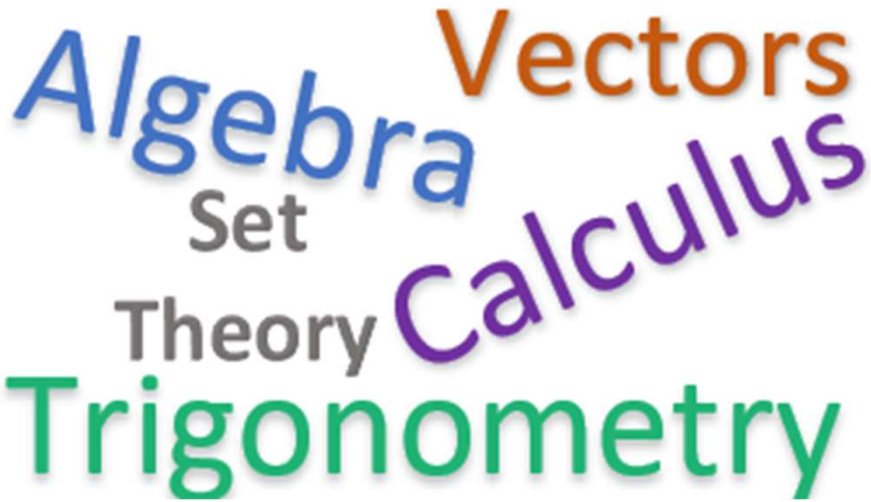
Crainix Understanding Advanced Maths JV1

Graphs are done in Desmos [Desmos | Graphing Calculator]





Learning happens best when:

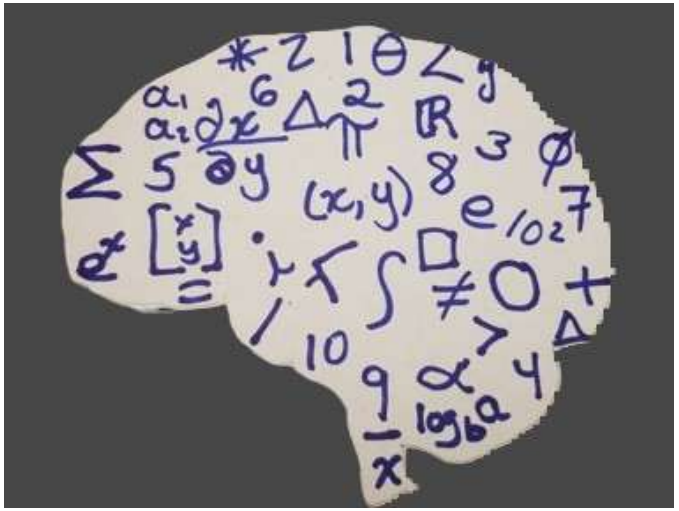
-there is learner choice and autonomy

-learning is fun and engaging

-learners can get into Flow

-it is a Just Right Challenge

-the focus is on learning rather than testing



What else facilitates learning for you?

All brains are different, so we all learn differently.

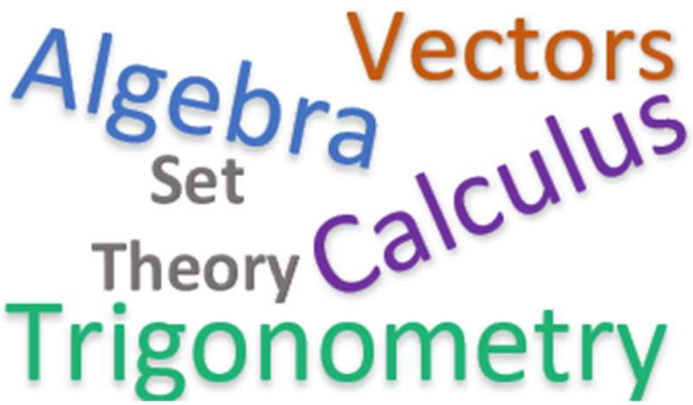
Please pay attention to how you learn.

Do you have any known specific learning differences or do you notice any as you are doing this program?

If so, let us know so we can improve the program to make it more inclusive.

Do you see different colors here?

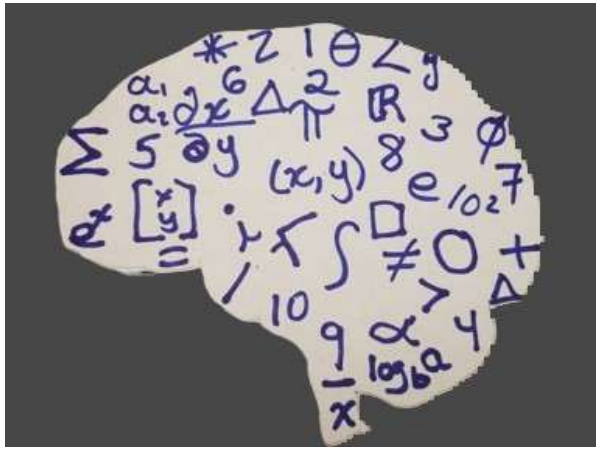
Color blindness – Wikipedia



We use competency-based education.

Learners decide how they want to be assessed.

How do you want to show what you know?

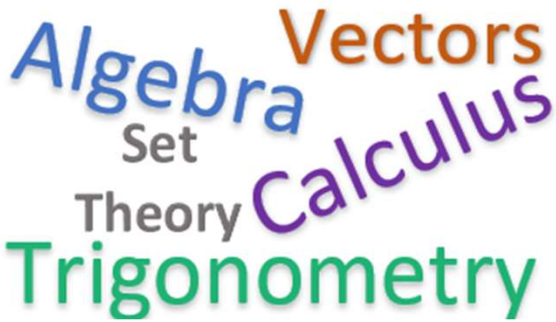


- Do practice problems?
- Create a portfolio?
- Make art or videos?
- Apply it for other subjects or projects?
- Program a computer to crunch numbers?
- Share your knowledge with someone else?
- Ask and answer questions as we go with verbal feedback?
- Other ideas?

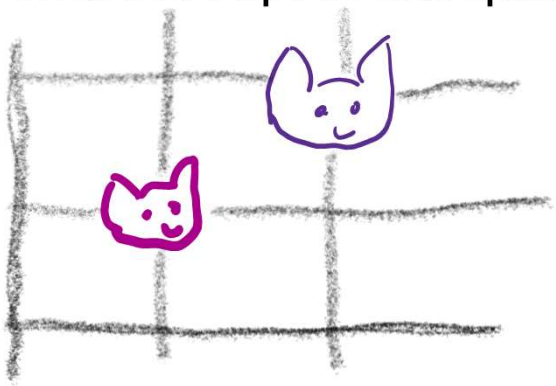
Crainix Understanding Advanced Maths

Understanding the concepts, rather than memorizing facts or focusing on calculations.

Advanced Maths using the plural or British way of saying mathematics to be inclusive of set theory, algebra, trigonometry, calculus, statistics, data science and more maths.



Numbers represent quantity, measurement, and location.



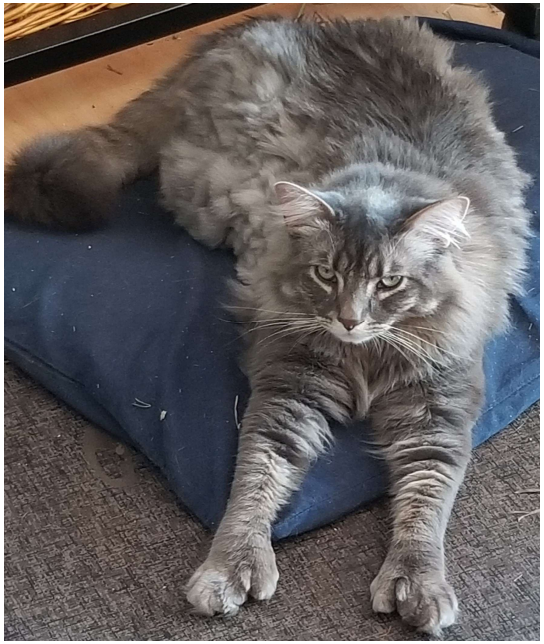
Symbols represent what we do with the numbers, called operations. Sometimes symbols represent numbers.

$1+1$ $\quad$ $2-1$ $\quad$ $\sqrt{2}$

$x-1$ $\quad$ π

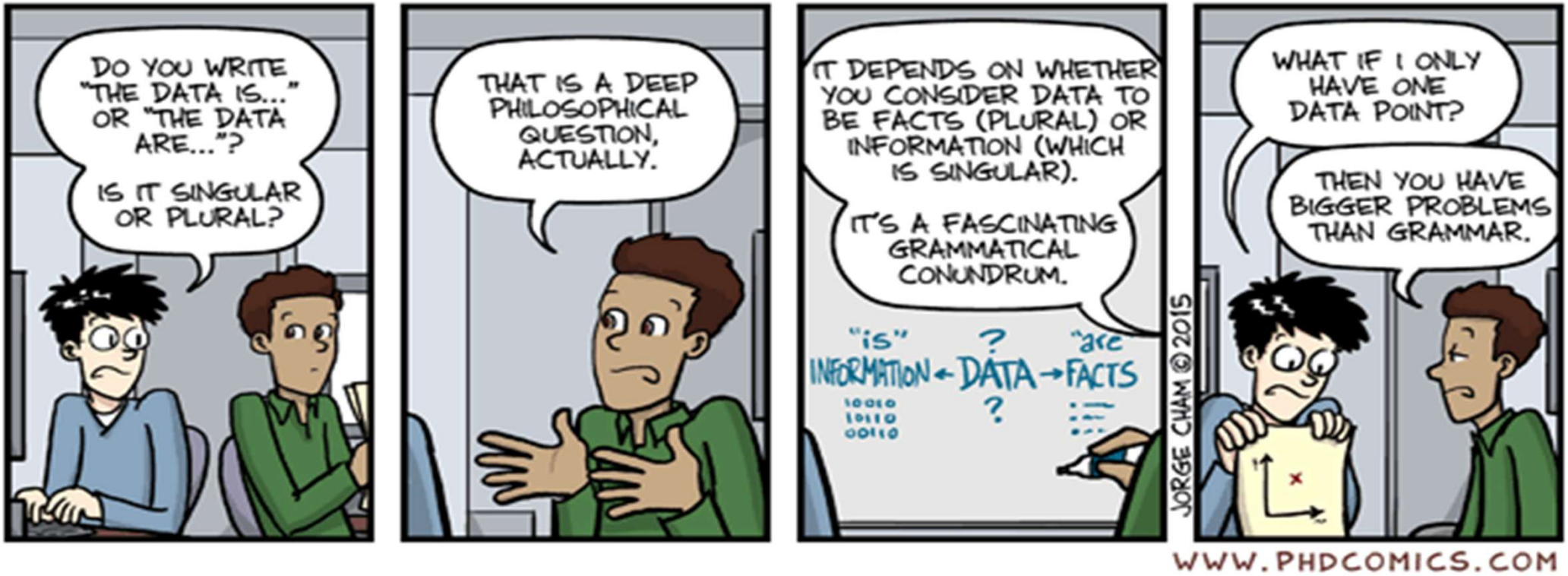
Data is the information that we collect or measure.

Doc cat grey 8 digits on each front paw

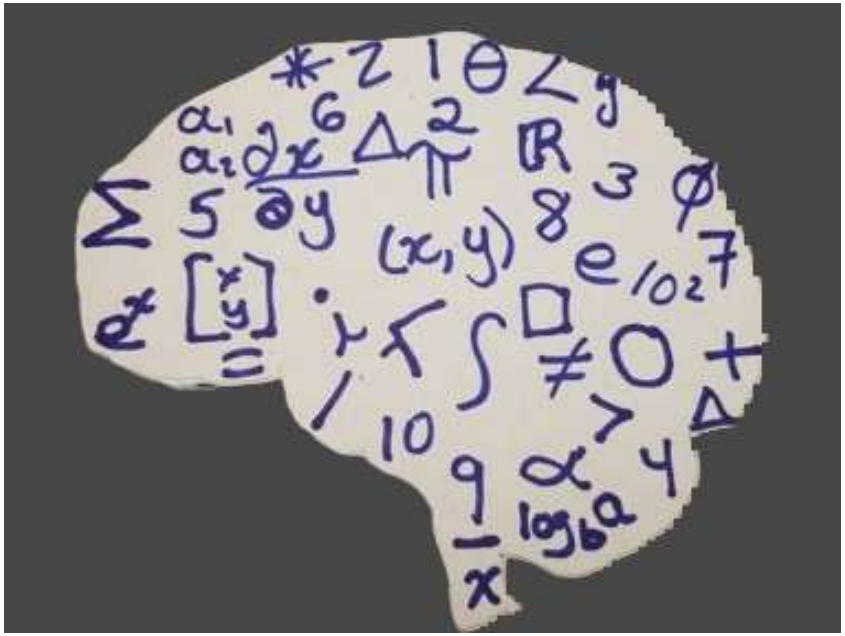


Data Data - Wikipedia

PHD Comics: A Grammatical Conundrum



Algebra is a basis for communicating and thinking about math. It uses symbols to represent numbers or data and there are also symbols for operations that can be done on the numbers or data.



We can use algebra to work with the data.

Concrete reasoning is when you are thinking about a particular object or example. It can be something tangible that you can touch, count or measure.

1 cat + 1 cat = 2 cats

Abstract reasoning is when you generalize to have it represent anything.

$n_1 + n_2 = S$

Some number + another number = a sum

Algebra lets us do abstract reasoning about situations that involve numbers.

Algebra uses variables, which are letters that represent numbers.

In the US, we use English and Greek letters most commonly for variables.

x, y and n

Using variables, we can go from concrete to abstract reasoning.

$1 + 1 = 2$ $\qquad$ $x + 1 = 2$

$3 + 2 = 5$ $\qquad$ $x + 2 = 5$

input number + a number = output number $\qquad$ $x + n = y$

These are called algebraic expressions. They describe what you are doing using the language of algebra. Here we are taking a number represented by the letter x and adding another number to it.

$x+1$ $x+2$

$x+n$

These are called algebraic equations. Do you notice that they have an equal sign? That is why we say equation.

$x+1=2$

$x+2=5$

$x+n=y$

Teaching Assistants for the program:

Socrates visits from Canada.

Doc, short for Doctor Strange.

Aury, short for Aurum because she is golden.